

CrabTransformer Training Report

Scenario: REAL | Loss: Tweedie | Lag: 5 Years | Current Spawner: Yes

— Training Configuration —

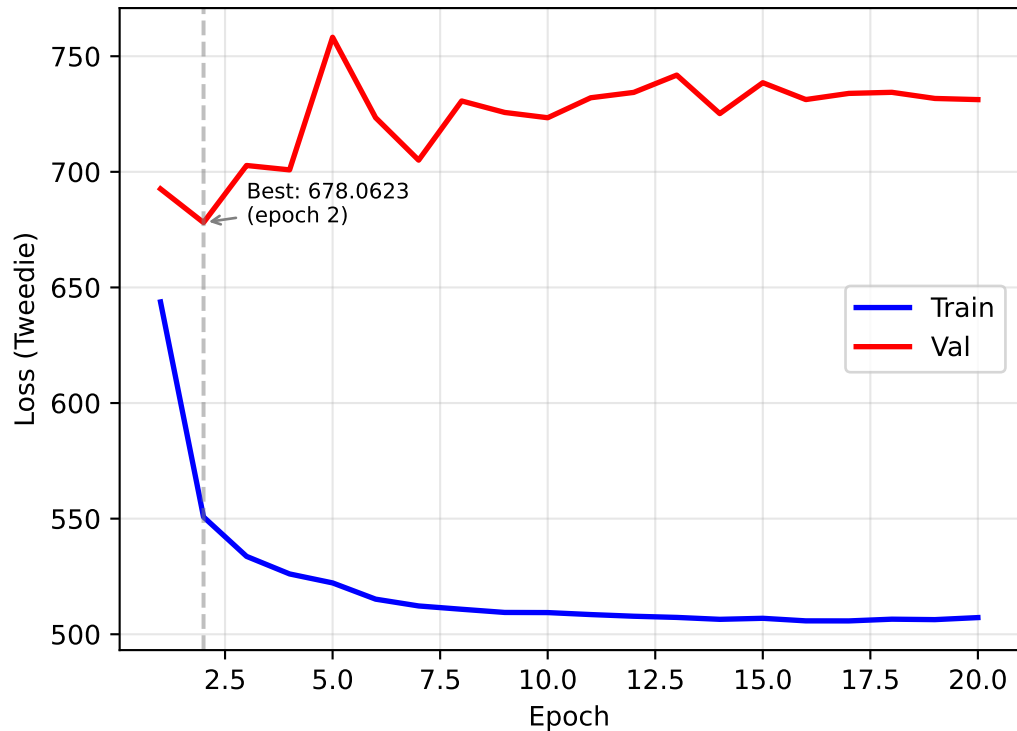
epochs_run: 20 best_val_epoch: 2
best_val_loss: 678.0623 final_train_loss: 507.2114
bias_correction (MSE): 1.0000
include_current_spawner: True

— Model Architecture —

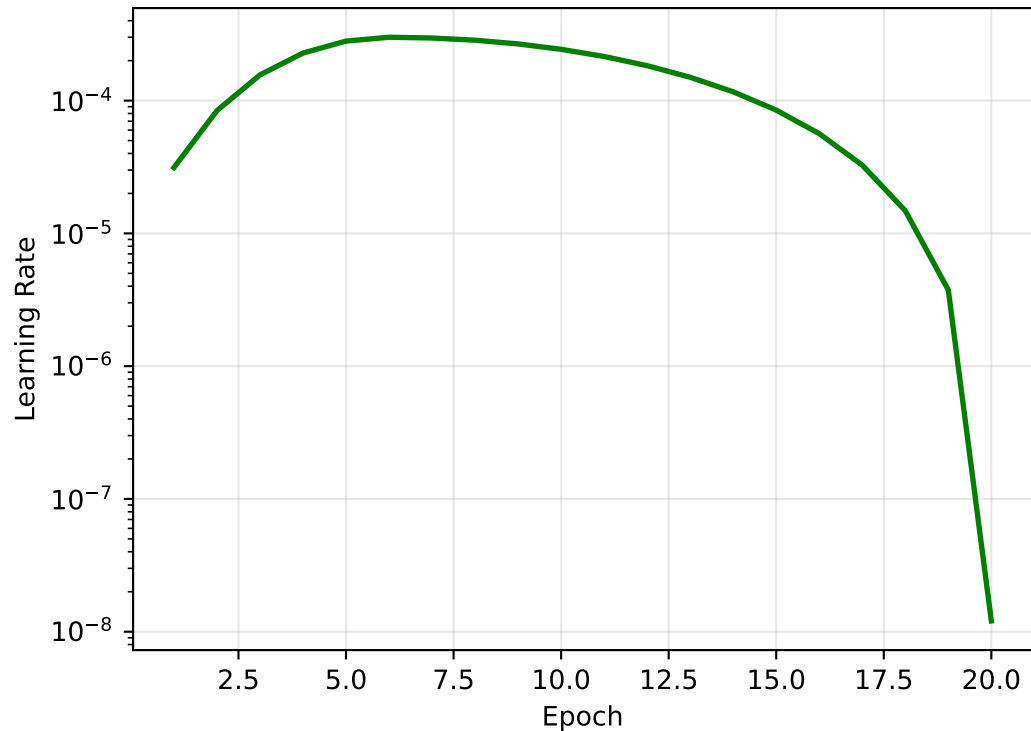
embed_dim: 128 num_heads: 8 num_layers: 6
d_ff: 512 dropout: 0.1 patch_size: 5
train/val/test years: 21/6/4

Training Curves: REAL | Lag 5 | Tweedie | w/ Spawner(t)

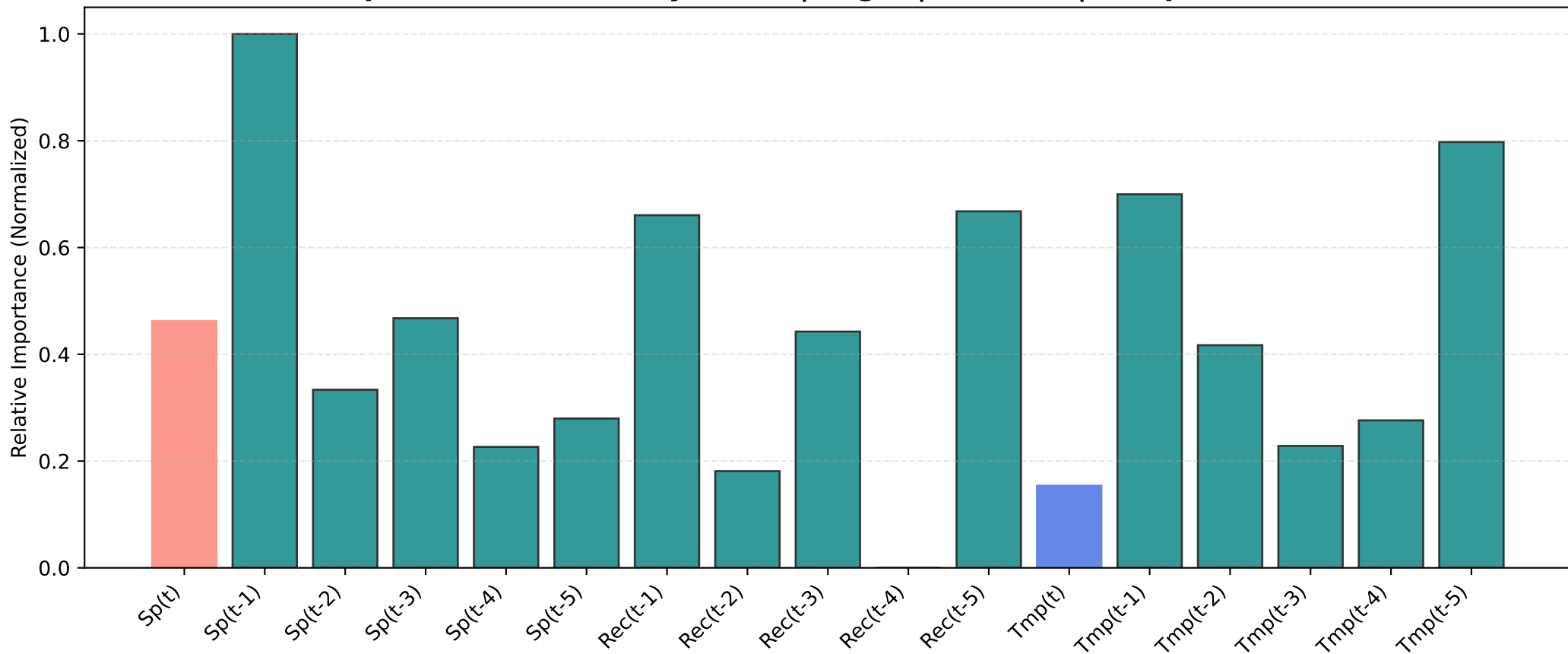
Training & Validation Loss



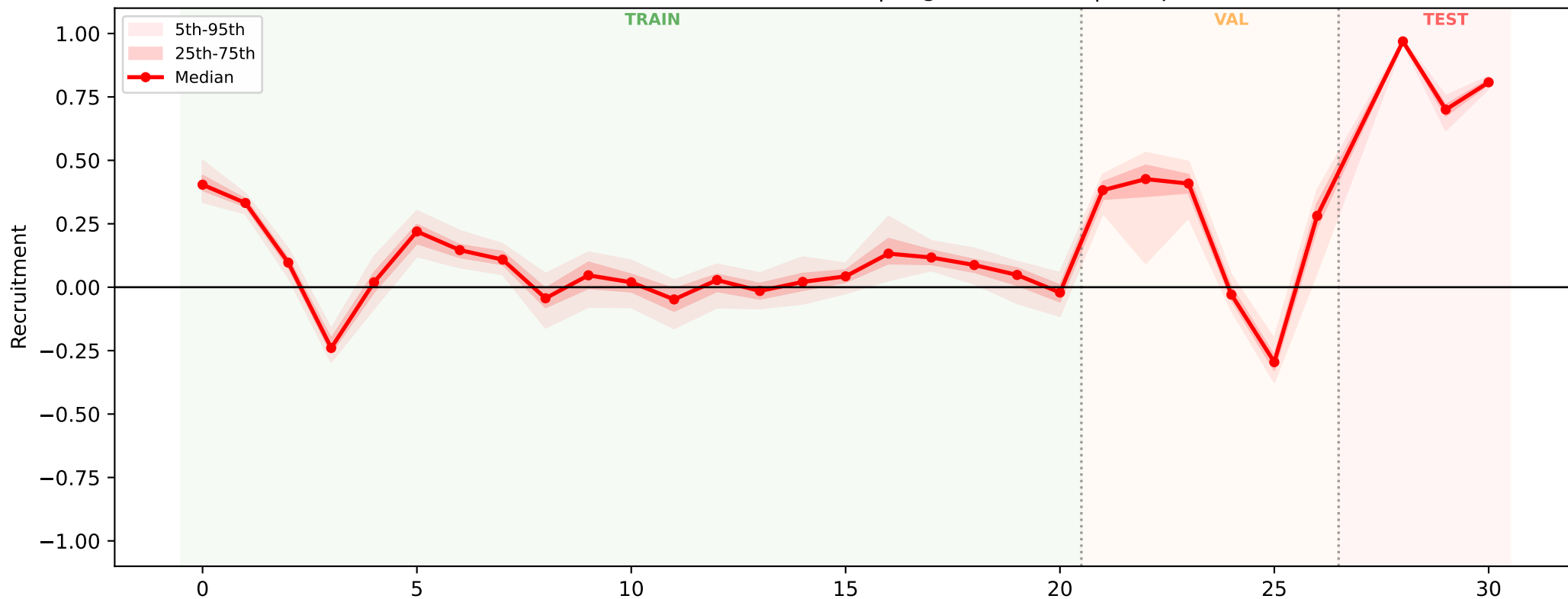
OneCycleLR Schedule



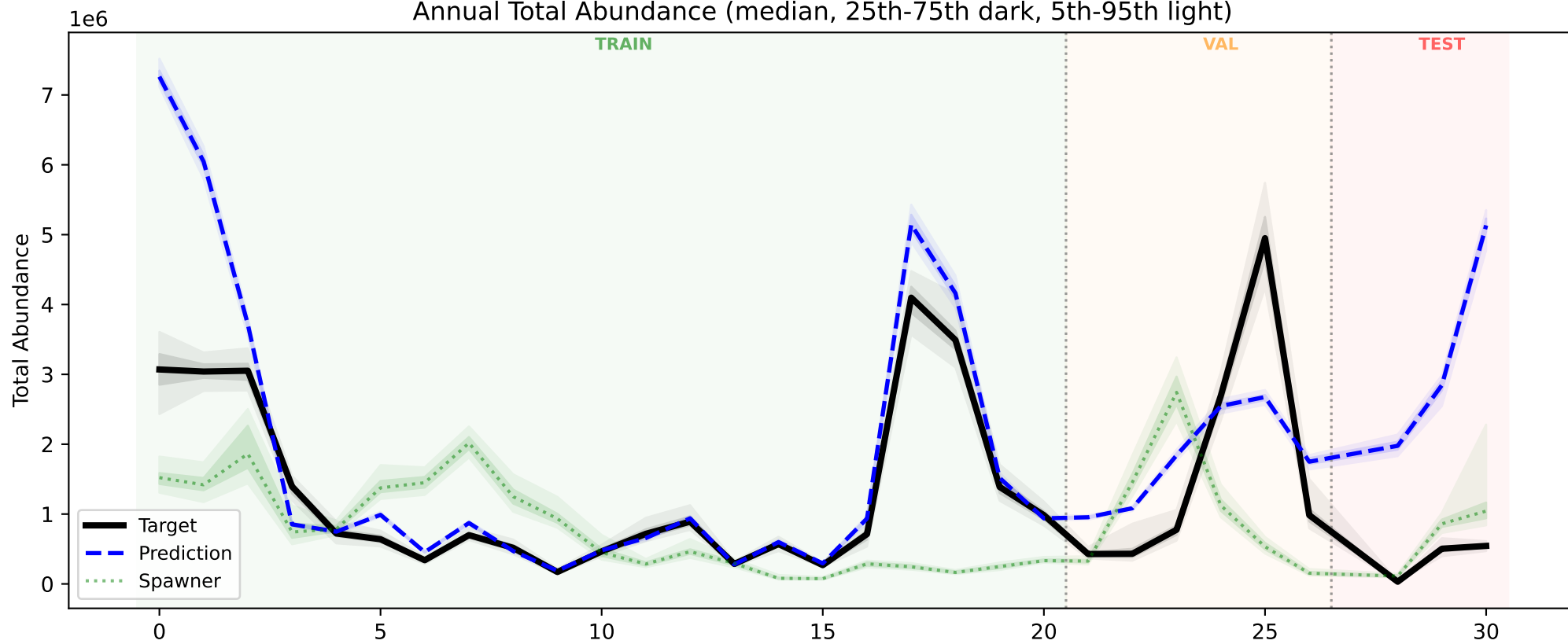
Input Channel Priority: REAL | Lag 5 | Tweedie | w/ Spawner(t)



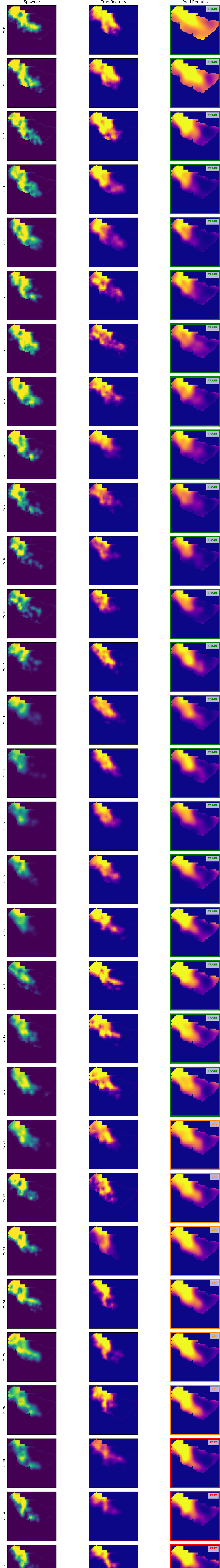
Annual Recruitment Deviations: REAL | Lag 5 (Tweedie) | w/ Spawner(t)



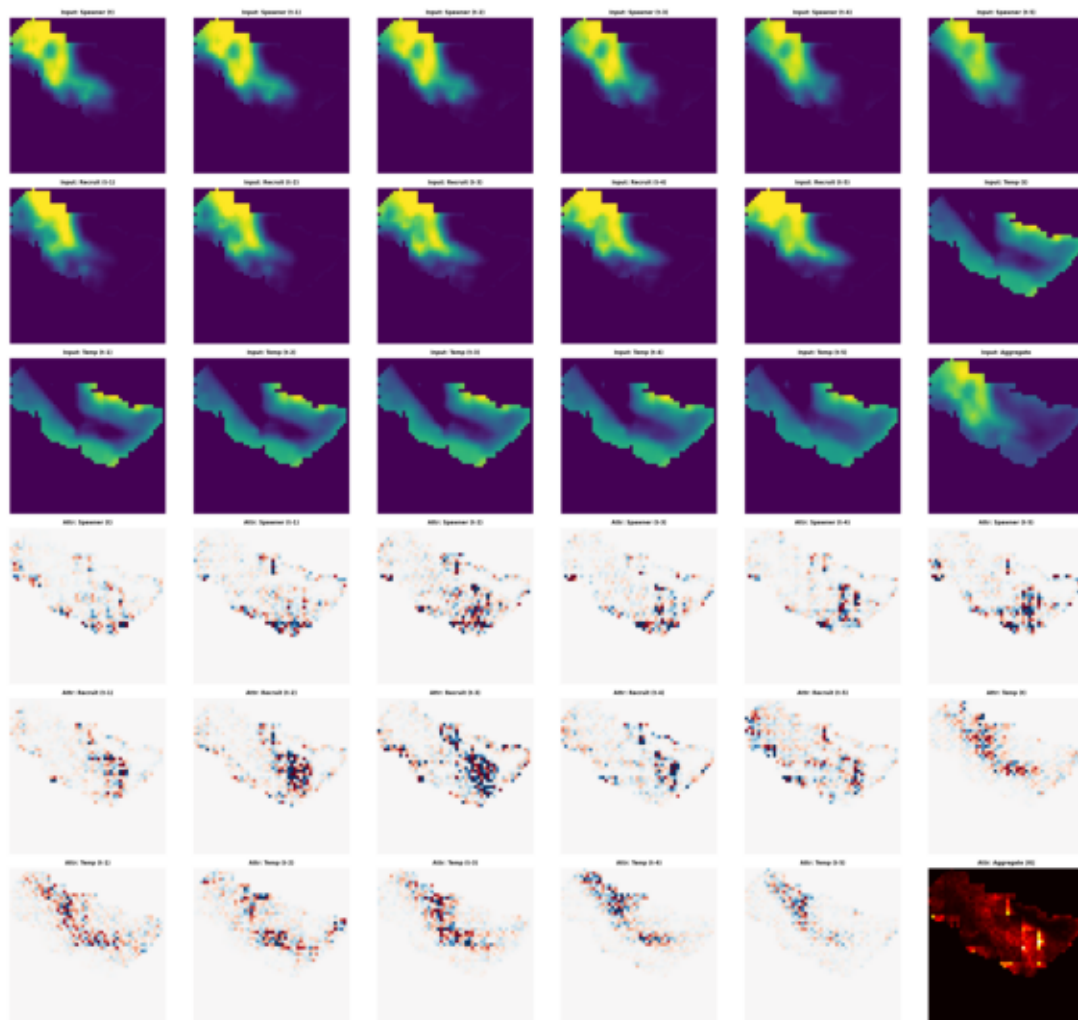
Annual Total Abundance (median, 25th-75th dark, 5th-95th light)



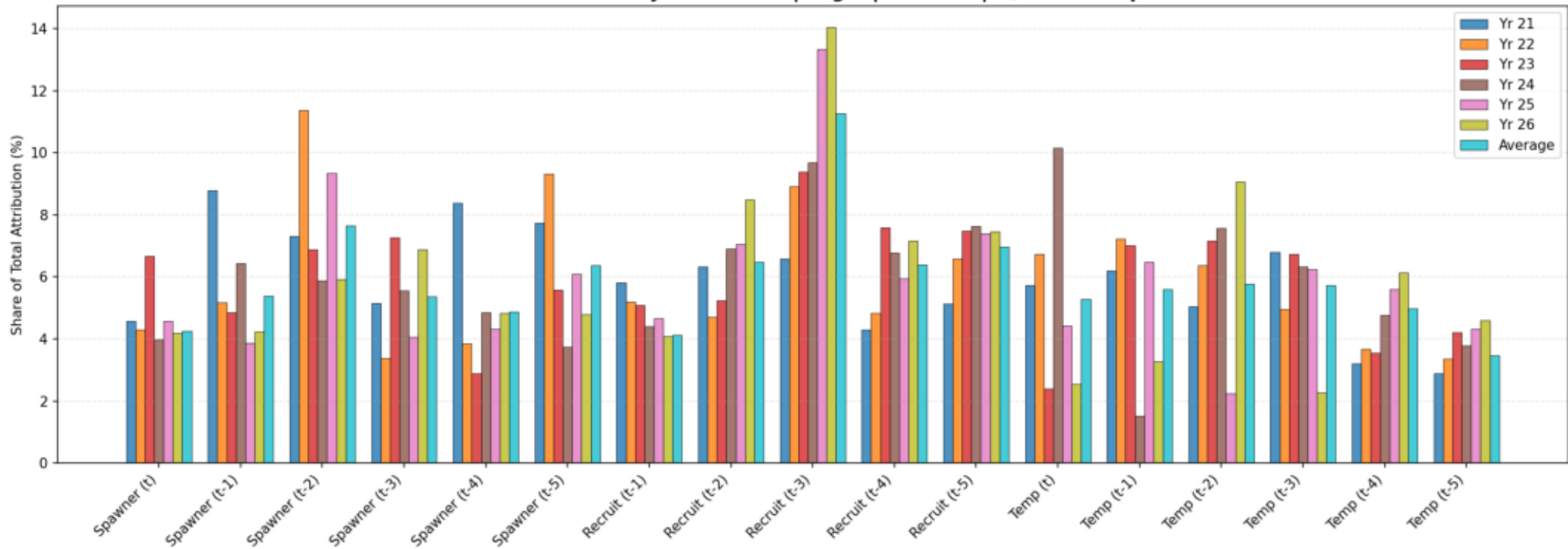
Spatial Trajectory: REAL | Lag 5 | Tweedie | w/ Spawner(t)



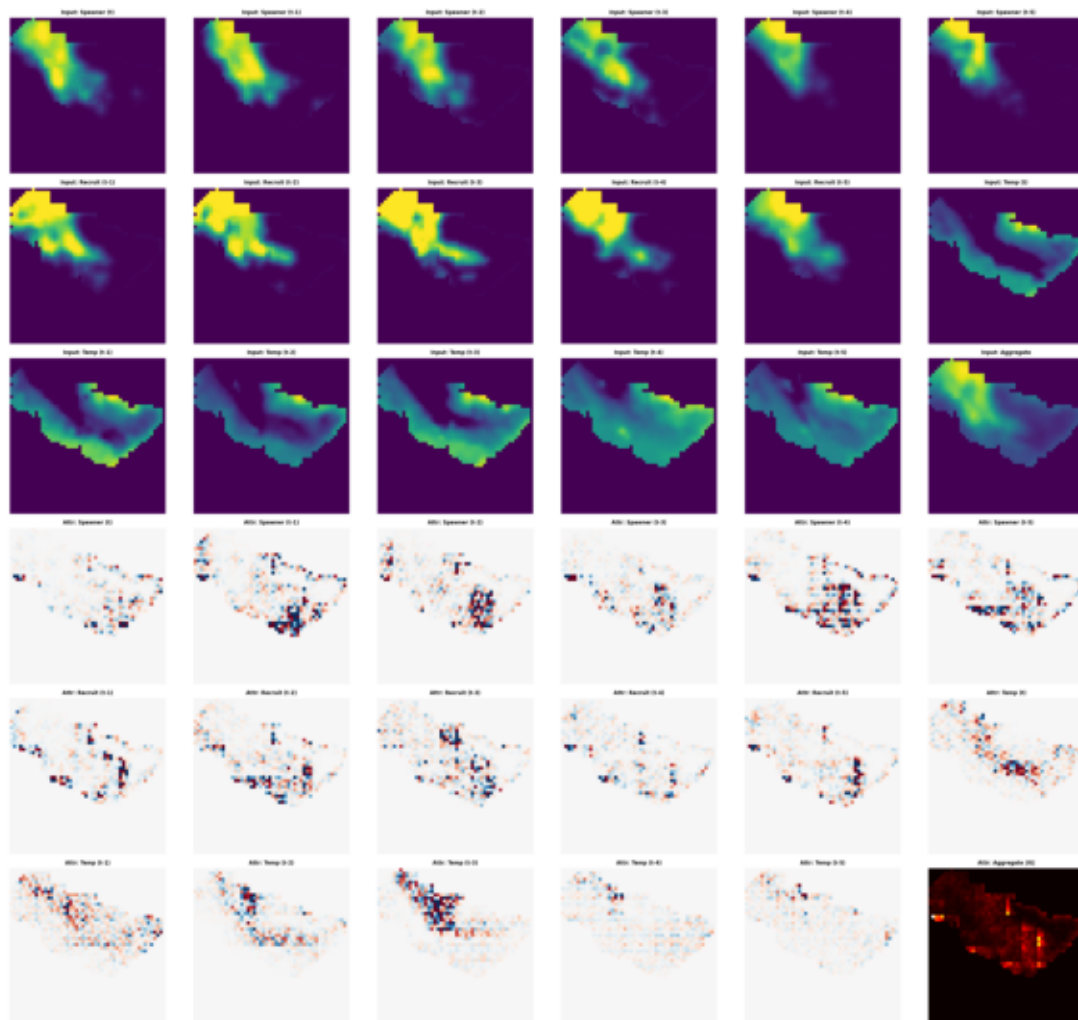
Year AVG — REAL | Lag 5 | Tweedie | w/ Current Spawner
 Top: Mean Input | Bottom: Attribution (red=+recruit, blue=-recruit)



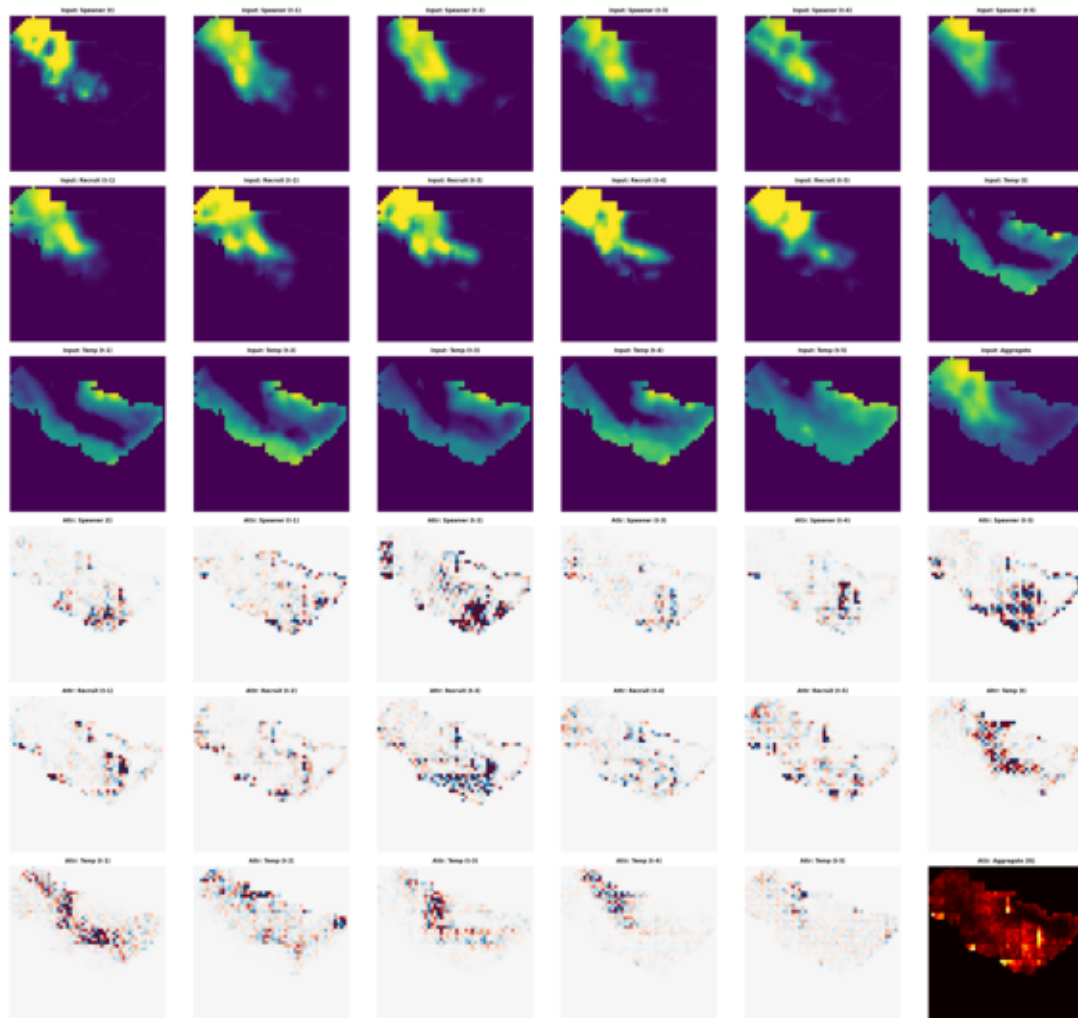
Channel Attribution by Year: REAL | Lag 5 | Tweedie | w/ Current Spawner



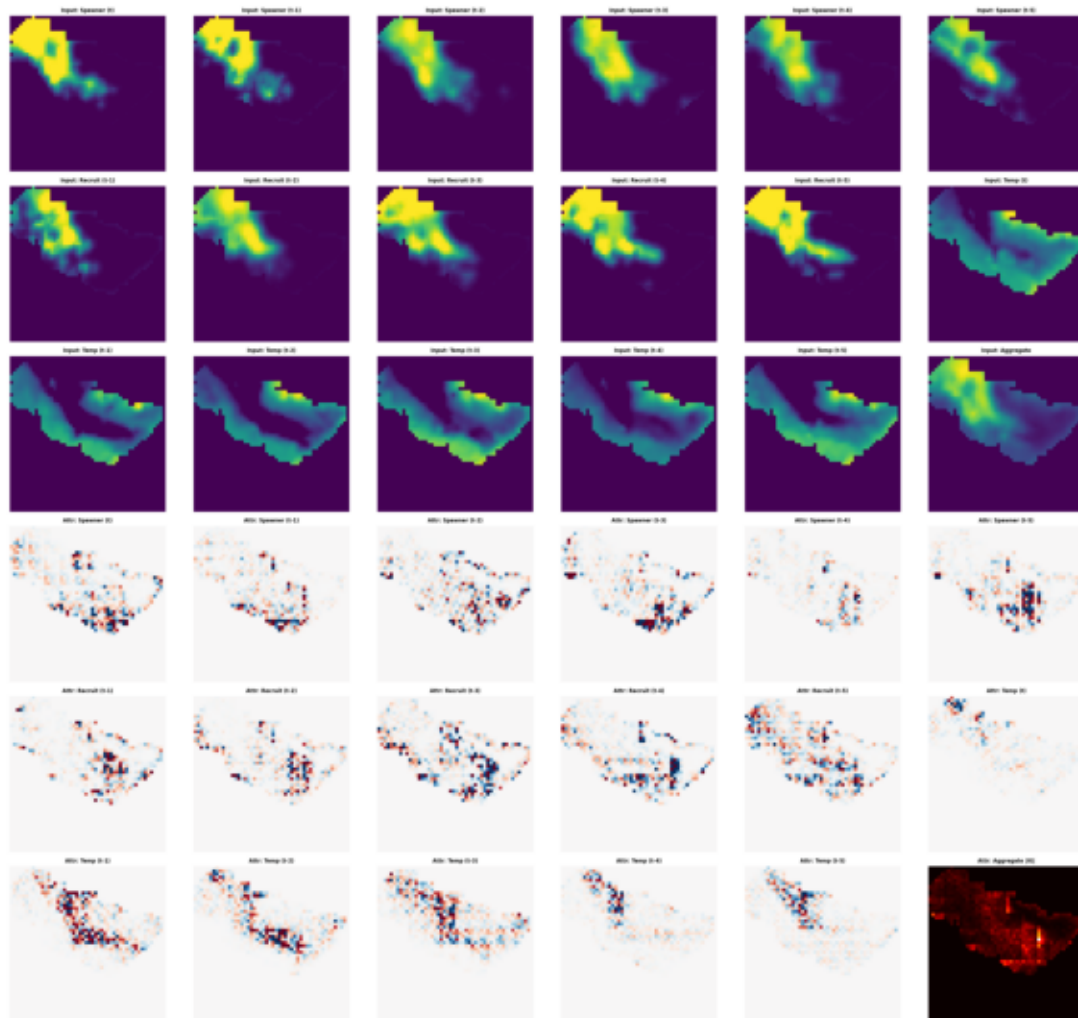
Year 21 — REAL | Lag 5 | Tweedie | w/ Current Spawner
 Top: Mean Input | Bottom: Attribution (red=+recruit, blue=-recruit)



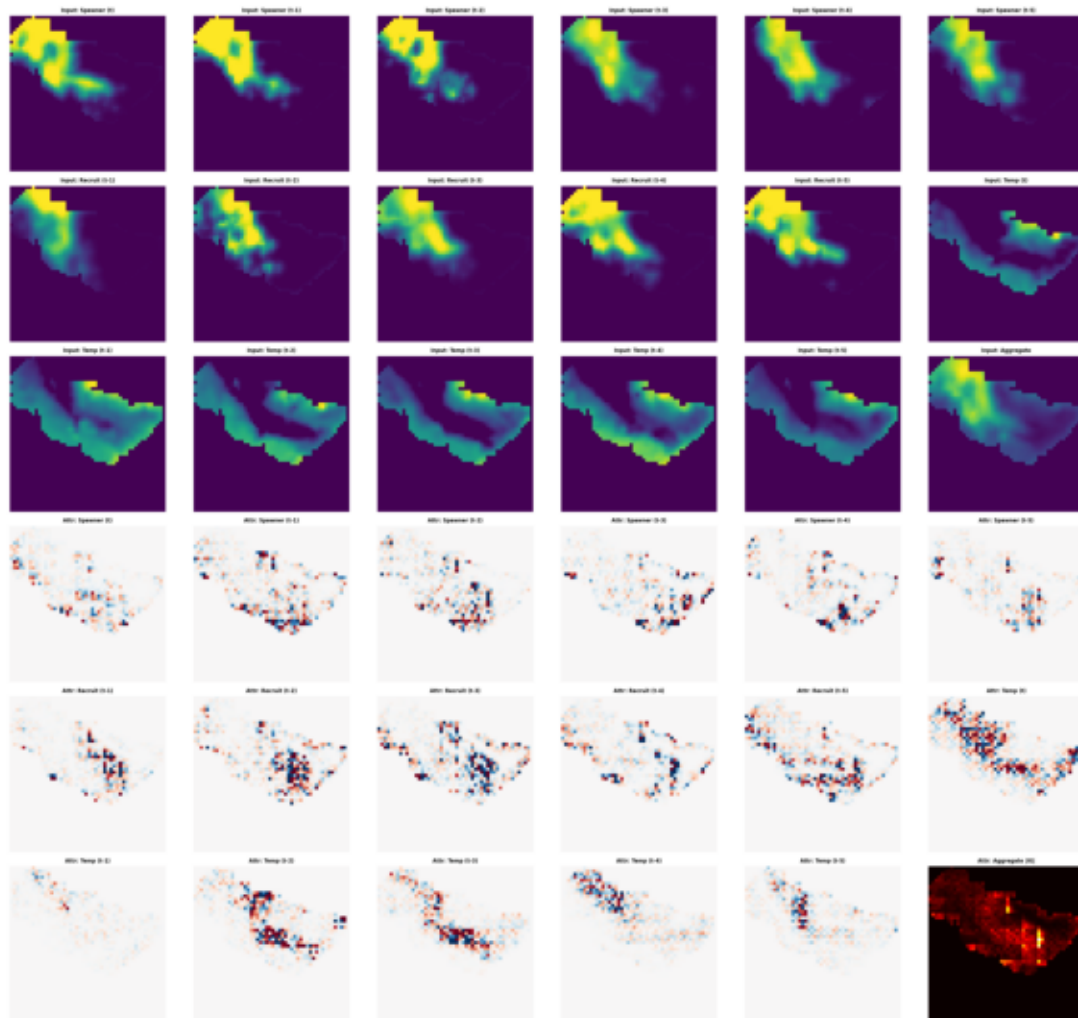
Year 22 -- REAL | Lag 5 | Tweedie | w/ Current Spawner
 Top: Mean Input | Bottom: Attribution (red=+recruit, blue=-recruit)



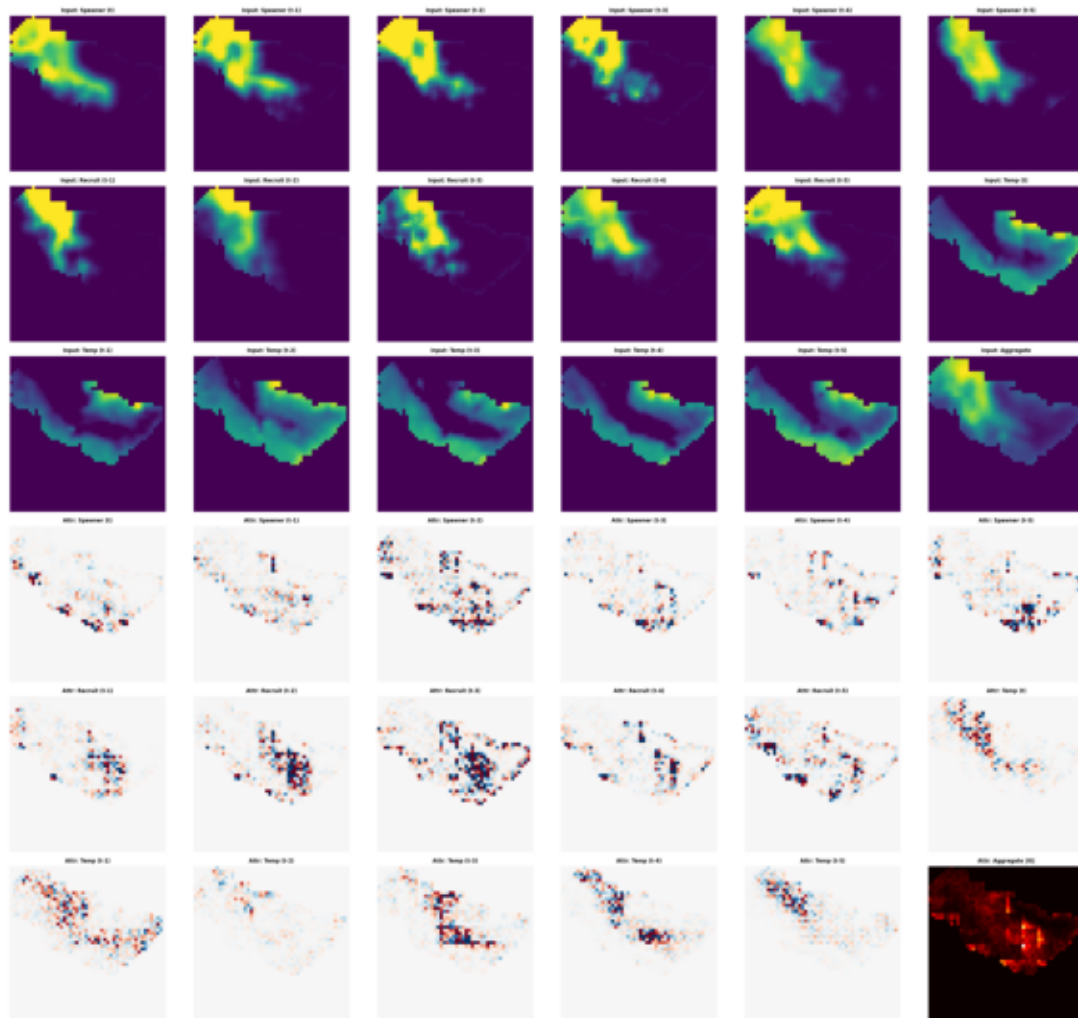
Year 23 -- REAL | Lag 5 | Tweedie | w/ Current Spawner
 Top: Mean Input | Bottom: Attribution (red=+recruit, blue=-recruit)



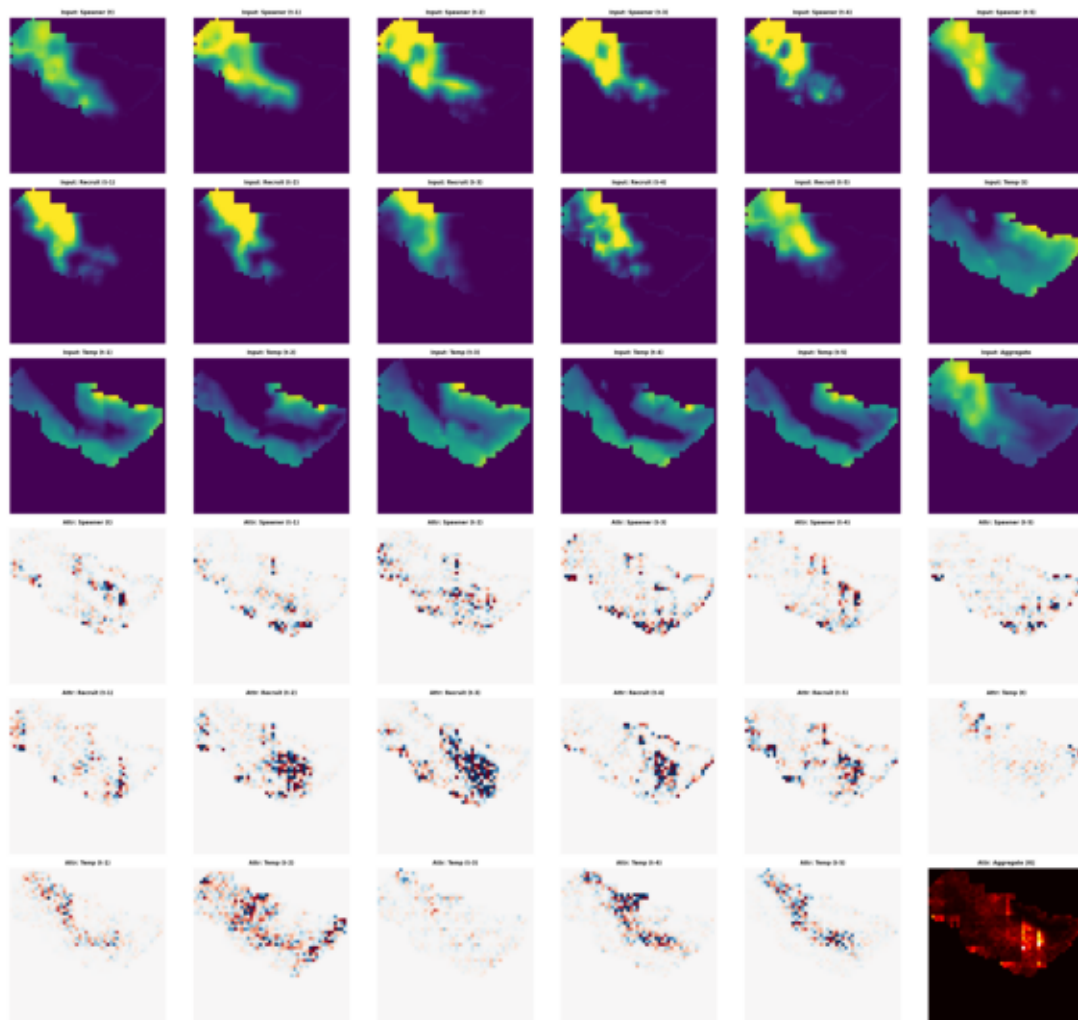
Year 24 — REAL | Lag 5 | Tweedie | w/ Current Spawner
 Top: Mean Input | Bottom: Attribution (red=+recruit, blue=-recruit)



Year 25 -- REAL | Lag 5 | Tweedie | w/ Current Spawner
 Top: Mean Input | Bottom: Attribution (red=+recruit, blue=-recruit)



Year 26 -- REAL | Lag 5 | Tweedie | w/ Current Spawner
 Top: Mean Input | Bottom: Attribution (red=+recruit, blue=-recruit)



Summary Statistics: REAL | Lag 5 | Tweedie | w/ Spawner(t)

Phase	Pred_Mean	Tgt_Mean	Pred_SD	Pred_P50	Pred_P95	Abundance_Capture	MAE	Spearman
TRAIN	2018.3	1480.3	7411.3	15.4	10589.6	1.363	1028.8	0.868
VAL	2038.0	1950.8	4667.7	59.6	11302.2	1.045	2036.3	0.794
TEST	3732.2	406.3	9533.7	18.0	21428.3	9.186	3363.5	0.754

Detailed Metrics (R-style): REAL | Lag 5 | Tweedie | w/ Spawner(t)

Phase	RMSE	RMSE_unbias	Bias	Bias_%	Spearman	Pearson	Pct_Error	Zero_F1	TopK_Acc	TopK_Jaccard	Quant_Err	Skill_MSE	Skill_MAE
TRAIN	2471.3	2361.8	538.0	3.4%	0.874	0.897	19.8%	0.804	0.785	0.653	0.96	0.665	0.628
VAL	5559.2	5429.6	87.1	7.8%	0.798	0.693	66.1%	0.735	0.693	0.544	1.63	-0.052	0.193
TEST	8344.9	7649.6	3325.9	16.7%	0.761	0.681	2593.7%	0.773	0.777	0.643	1.56	-917.018	-15.570